Universal Projector Geometry from Electrical Measurements: A Coordinate-Free Framework for Transport, Persistence, and Emergent Structure

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Abstract

We present a unified, operator-first framework in which macroscopic geometric structure emerges from the continuous tracking of persistent spectral projection bundles on stratified configuration spaces, bypassing the conventional requirement of an a priori background metric tensor. The electric side of the framework rests on a single root operator: Dirac's, whose algebra, paired spectrum, and integer invariants organize every observable developed here. Module I (Geometric Admissibility) develops the intrinsic Hilbert–Schmidt quantum metric of the Grassmannian and a canonical redistribution operator $F(\Omega, \Pi)$ quantifying the incompatibility of structural change with the current transport decomposition; its algebraic theorems are gated numerically, and its twisted torus anchor is carried onto a finite noncommutative torus, where the integer record acquires an explicit domain of validity and an amplitude floor. Module II (Topological Continuation) uses character varieties and the A-polynomial of the figure-eight knot complement as a structural laboratory for global continuability, with the integer transport record carried by a verified winding invariant whose self-adjoint ancestor is the Levine–Tristram signature. Module III (Covariance Realization) defines every intrinsic observable — principal angles, transport capacity, chirality, rigidity, persistence, refraction, and effective dimension — with explicit estimators, realizes the framework on published Bloch-state tomography of a Bose–Einstein condensate where the winding currency reproduces the measured Chern number, and specifies the independence protocol under which the interface refraction test becomes falsifiable on the instrument's archived data. Every claim carries an explicit grade; verification scripts are included as the proof of record.

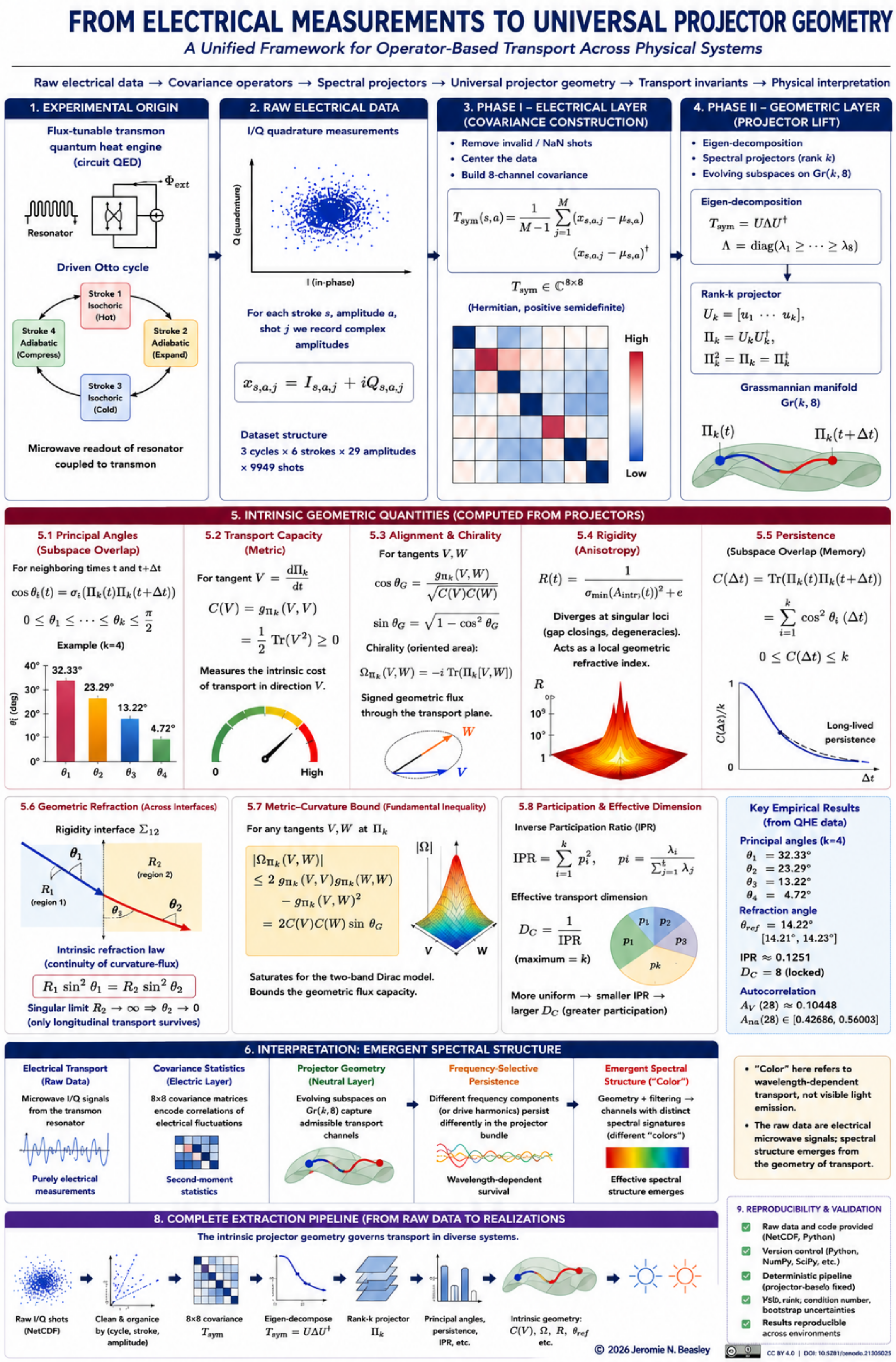

Figure 1: From electrical measurements to intrinsic transport geometry. Raw electrical data → covariance operators → spectral projectors → universal projector geometry → transport invariants → physical interpretation.

1 What is claimed, and at what grade

Every claim in this paper carries one of five grades, never silently promoted: [T] exact/derived (closed forms and identities); [V] verified (gated numerical checks, scripts in Appendix B); [C] cited (established literature, attributed); [D] data-reported (empirical values from the instrument pipeline, quoted with the protocol that must accompany any re-run); [H] hypothesis, protocol, or interpretation. Where a chain mixes grades, the weakest link is named.

1 Framework Overview: the Redistribution Operator

The framework begins with raw electrical measurements and extracts coordinate-free geometric structure through a two-phase pipeline. In Phase I (data-driven), covariance matrices are formed from single-shot I/Q quadrature data and decomposed into rank-k spectral projectors $\Pi_k(t)$ on the Grassmannian. In Phase II (framework-dependent), intrinsic geometric quantities are computed from these projectors (§6).

To describe how admissible subspaces change under structural evolution we introduce a second fundamental operator.

Definition 1.1 (Structural Transition Generator)**.** A structural transition generator Ω is an operator on the underlying Hilbert space representing an infinitesimal or discrete structural deformation of the current spectral decomposition. Unlike the projector Π, which identifies the presently admissible transport subspace, Ω is not required to preserve that decomposition and may transfer spectral weight between the projector and its orthogonal complement. Ω may be realized by weighted transport operators, Jacobi stiffness operators, Grassmannian connection generators, experimentally inferred covariance deformations, or effective Hamiltonian perturbations; the development depends only on the algebra relating Ω and Π (Appendix A).

Given such a generator, the redistribution operator is

$$F(\Omega, \Pi) \;=\; (I - \Pi)\,\Omega\,\Pi \;+\; \Pi\,\Omega\,(I - \Pi). \tag{1}$$

Theorem 1.2 (Equivalent forms and structural properties; [T]+[V])**.** *$F(\Omega, \Pi)$ admits the equivalent representations $F = [\Omega, \Pi]\Pi + \Pi[\Pi, \Omega]$ and $F = \Omega\Pi + \Pi\Omega - 2\Pi\Omega\Pi$. Moreover F is traceless and purely off-diagonal with respect to the decomposition induced by Π: $\Pi F \Pi = 0$ and $(I-\Pi)F(I-\Pi) = 0$.*

Theorem 1.3 (Compatibility criterion; [T]+[V])**.** *$F(\Omega, \Pi) = 0$ if and only if $[\Omega, \Pi] = 0$.*

Corollary 1.4. *The scalar transport intensity $T(\Omega, \Pi) := \|F(\Omega, \Pi)\|_F$ is non-negative, vanishes precisely when the generator commutes with the projector, and is invariant under unitary changes of basis.*

All identities of Theorems 1.2–1.3 are gated numerically: equivalent forms, tracelessness, and cross-subspace support hold over 200 random Hermitian-generator trials, and the constructive direction of the compatibility criterion holds to 9×10^{-16} (Appendix B).

Hypothesis 1 (Hysteresis under topological driving)**.** When a control parameter is driven forward and backward across a topological critical point at finite rate, the geometric quantities extracted from the evolving projectors (in particular d_{eff} and $C(\Delta t)$) exhibit history dependence, visible as a statistically significant nonzero area between forward and backward trajectories.

Hypothesis 2 (Correlations with transport intensity)**.** Larger $T(\Omega, \Pi)$ correlates with increased projector redistribution, larger changes in effective dimensionality, and stronger persistence effects across rigidity interfaces.

2 The Dirac Foundation of the Electric Side

The electric side of this framework has one root operator. Dirac's equation, in linearizing energy, produced three permanent structures: the σ-algebra, which every two-level and two-band system since has inherited; the paired $\pm$ spectrum, the first chiral spectrum in physics; and the electron itself, whose transport is the subject of every covariance matrix measured in this work. The framework's observables, currencies, and material realizations are organized as consequences of this single foundation.

The integer currencies. A Dirac operator carries its topology in integers: zero modes counted by the index; spectral asymmetry measured by ρ-invariants, which on knot-complement boundaries take the form of the Levine–Tristram signature $\sigma_K : S^1 \to \mathbb{Z}$ — an integer step function whose plateaus jump exactly at the Alexander roots on the holonomy circle [T]; and the winding of the mass vector, whose loop integrals are exact integers [V]. These three integers — index, asymmetry, winding — are the currencies in which every topological claim of this framework is denominated. Modern index theory realizes the same structure in topological matter, where phases are classified by η-invariants and spectral flow under Atiyah–Patodi–Singer boundary conditions [C].

The geometric identity. Every two-band Hamiltonian $h(k) \cdot \sigma$ is a Dirac Hamiltonian with a momentum-dependent mass vector, and for its band projectors the metric–curvature bound of §6.2 holds with equality — an exact identity of the Dirac class [T], verified to machine precision and realized on measured data in §6.5. Multi-band systems obey the same bound as an inequality; its saturation is the signature of an effectively Dirac sector.

The material ladder. The Dirac operator's laboratory realizations span the platforms of this program. Graphene realizes the massless operator: the Landau ladder $E_n = v_F\sqrt{2\hbar eBn}$ [V: $\sqrt{n}$ ratios exact to 10^{-6} on the ladder operator], with a single $n = 0$ level seating the parity anomaly and a Berry phase of exactly π around the cone [V], anchored by the measured $v_F = 1.10 \times 10^6$ m/s giving $E_1(10\,\mathrm{T}) = 126$ meV, the scale of the observed scanning-tunneling ladder [C]. The charge-neutrality point is the Dirac point occupied by a quantum-critical electron fluid whose viscous flow has been imaged directly [C]; the quantum-anomalous-Hall lattice bands realized in cold-atom tomography are lattice Dirac operators with mass textures [C]; the nonlinear-Hall hotspots of the quantum-metric-dipole experiments are massive Dirac points [C]; and the transmon qubit is the bare σ-kernel — the Dirac algebra with the field stripped away, which is why covariance data from a driven transmon engine furnishes a faithful substrate for projector geometry.

The chiral spectral class. Spectra symmetric under $\lambda \to -\lambda$ form the Dirac operator's ensemble class, and the ledger functional $Z/M = \frac{1}{2}\int \rho(\lambda)\ln(1+\lambda^2)\,d\lambda$ is native to it: the half-Cauchy family gives $Z/M = \ln(1 + a)$ in closed form [T], and the chiral-Gaussian ensemble (Marchenko–Pastur singular-value law) matches its integral at finite size to 5×10^{-5} [V]. The constant $\pi^2/12$ is realized at a calibrated width in each family: $a^* = e^{\pi^2/12} - 1 = 1.276108$ (half-Cauchy) and $a^* = 2.665$ (chiral Gaussian) [T+V].

Scope. Non-normal advective and dissipative flow operators belong to a different class and are outside this paper. What extends beyond the Dirac family is not the operator but the currencies — chirality, winding, spectral flow — each an integer record carried by a spectral object through a wall. The electric side is where those currencies are minted, and they are minted by Dirac's operator.

3 Global Admissibility and Axiomatic Bulk Structure

Definition 3.1 (Admissible bulk)**.** A compact, orientable 3-manifold X with boundary ∂X is an admissible bulk if: (1) each component of ∂X supports a first-order elliptic boundary operator whose holonomy family carries the integer winding record of §4.1; (2) the boundary data arise as the restriction of a global bulk structure and flat $U(1)$ connection on X; (3) $\pi_1(\partial X) \hookrightarrow \pi_1(X)$ is injective up to conjugacy (incompressible boundary); (4) Spectral rigidity: the integer record is locally constant in the holonomy parameters, with jumps confined to a discrete wall set; (5) Minimality: the total record satisfies $|n_{\text{total}}| = 1$.

4 Module I: Twisted Boundary Transport and Continuation Obstructions

4.1 The torus anchor and phase-transport calculus

Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ carry the twisted operator family

$$D_{\alpha,\beta} = i\sigma_1(\partial_x + 2\pi i\alpha) + i\sigma_2(\partial_y + 2\pi i\beta), \qquad \lambda^{\pm}_{m,n} = \pm 2\pi\sqrt{(m+\alpha)^2 + (n+\beta)^2}. \tag{2}$$

With the complex mode coordinate $Z_{m,n} = 2\pi[(m+\alpha) + i(n+\beta)]$, the positive-sector projector is $P_{m,n} = \frac{1}{2}\begin{pmatrix} 1 & \bar{Z}/|Z| \\ Z/|Z| & 1 \end{pmatrix}$ and the eigenspace orientation is the phase $e^{i\Phi_{m,n}} = Z/|Z|$.

Theorem 4.1 (Phase keeps the record; [T]+[V])**.** *Along any smooth deformation $(\alpha(t), \beta(t))$, the scalar eigenvalue $\lambda = |Z|$ compresses directional input into an orientation-blind magnitude, while the phase obeys the transport law*

$$\frac{d\Phi_{m,n}}{dt} = \frac{(m+\alpha)\dot{\beta} - (n+\beta)\dot{\alpha}}{(m+\alpha)^2 + (n+\beta)^2}, \tag{3}$$

a gauge-invariant angular velocity on the holonomy torus. Loop integrals of (3) around the spectral zero are exact integers: winding 0 for non-enclosing loops, $+1$ per enclosure, $+2$ for a double loop (verified to 10^{-6}, Appendix B).

The currency. The spectrum of (2) is symmetric under $\lambda \to -\lambda$ by the exact chiral symmetry $\{\sigma_3, D_{\alpha,\beta}\} = 0$, so the spectral-asymmetry invariant η vanishes identically on this family; the integer content of transport resides in the winding record of Theorem 4.1. Its self-adjoint knot-theoretic ancestor is the Levine–Tristram signature $\sigma_K : S^1 \to \mathbb{Z}$, an integer step function jumping exactly at the Alexander roots on the holonomy circle (trefoil: plateau -2 between holonomies $1/6$ and $5/6$; the amphichiral 4_1: $\sigma \equiv 0$) [T].

4.2 Local neutrality and global obstruction lower bounds

The obstruction curvature is formalized via the regularized quartic potential $U_C(T) = (T^2 - a^2 I_N)^2 + \epsilon I_N$ with curvature operator $R_C(T) = 4(3T^2 - a^2 I_N)$; local neutral sectors occur at $T_* = \pm(a/\sqrt{3})I_N$, where $\ker R_C(T_*) \neq 0$. The global obstruction along a normalized path is $O(\gamma) = \int_{-a}^{a} \frac{1}{N}\mathrm{Tr}\, U_C(T(t))\, dt$.

Theorem 4.2 (Persistence of the global bound; [T], commuting reduction)**.** *Local neutral sectors do not permit the global obstruction to vanish along a continuous deformation. For the commuting reduction $T(t) = tI_N$,*

$$O_{\min} \;\geq\; 2\int_0^a (t^4 - 2a^2t^2 + a^4 + \epsilon)\, dt \;=\; \frac{16}{15}a^5 + 2a\epsilon \;>\; 0. \tag{4}$$

The fully non-commutative lower bound remains conjectural; this verification relies on the commuting reduction, and the grade is stated accordingly.

4.3 Interface gluing anomalies

When continuation patches A and B are glued along a boundary component Σ, the combined connection operator is $R_M = \begin{pmatrix} R_A & \epsilon_{B\Sigma} \\ \epsilon_{B\Sigma}^\dagger & R_B \end{pmatrix}$.

Assumption 1. $(I + R_A)$ and $(I + R_B)$ are strictly positive definite and $\epsilon_{B\Sigma}$ is bounded.

Theorem 4.3 (Interface anomaly; [T] under Assumption 1)**.** *Gluing under Assumption 1 yields a strictly positive non-local interface correction: by Schur decomposition and trace-log equivalence, $O(A \cup_\Sigma B) = O(A) + O(B) + \Delta_{\text{interface}}$ with*

$$\Delta_{\text{interface}} = \mathrm{Tr}\log\Big(I + \epsilon_{B\Sigma}^\dagger (I + R_A)^{-1} \epsilon_{B\Sigma} (I + R_B)^{-1}\Big) = \sum_k \log(1 + \epsilon^2 \mu_k) > 0, \tag{5}$$

with $\mu_k > 0$ the eigenvalues of the localized boundary product.

4.4 The noncommutative torus anchor

The torus family of §4.1 carries its holonomy in two parameters α, β that commute. Nothing in the winding currency requires them to. Replacing the two holonomy coordinates by two unitary operators that fail to commute by a scalar moves the same anchor onto a finite-dimensional non-commutative torus, and the integer record survives the move with an explicit domain of validity.

Definition 4.4 (Noncommutative anchor and its readouts)**.** Let $U, V \in \mathrm{U}(d)$ and set the relator

$$R \;=\; UVU^\dagger V^\dagger, \qquad \lambda_1, \ldots, \lambda_d \;=\; \mathrm{spec}(R). \tag{6}$$

The *defect*, the *branch guard*, and the *winding record* are

$$\delta = \frac{\|R - I\|_F^2}{d}, \qquad g = \min_j |\lambda_j + 1|, \qquad w(U, V) = \frac{1}{2\pi}\sum_{j=1}^{d} \arg \lambda_j, \tag{7}$$

with arg the principal branch. The anchor is *admissible* at tolerance τ when $g > \tau$.

The defect is the normalised Hilbert–Schmidt distance of the relator from the identity: the amplitude by which the two generators fail to commute. The winding is the phase that failure accumulates across the spectrum, and it is the same currency the smooth family of §4.1 carries around a spectral zero.

Theorem 4.5 (Integrality and invariance; [T]+[V]). *Let $g > 0$. Then (i) $w(U, V) \in \mathbb{Z}$, and $|w| \le d/2$; (ii) $w(QUQ^\dagger, QVQ^\dagger) = w(U, V)$ for every $Q \in \mathrm{U}(d)$; (iii) $w(e^{i\theta}U, e^{i\varphi}V) = w(U, V)$ for all $\theta, \varphi \in \mathbb{R}$; (iv) $w(U, V)$ is the winding number about the origin of the closed loop $t \mapsto \det\big((1-t)I + tR\big)$, $t \in [0, 1]$.*

Proof. R is a commutator of unitaries, so $\det R = 1$ and hence $\prod_j \lambda_j = 1$. Taking arguments, $\sum_j \arg \lambda_j \in 2\pi\mathbb{Z}$, which is (i); the bound follows from $|\arg \lambda_j| < \pi$. For (ii), conjugation carries R to $QRQ^\dagger$ and preserves the spectrum. For (iii), the scalars cancel identically inside the commutator. For (iv), $\det R = 1$ at $t = 1$ and $\det I = 1$ at $t = 0$, so the path closes; the identification of its winding with the summed principal arguments is the Exel–Loring formula [C], whose two formulations — K-theoretic and winding-number — were proved to agree. □

The determinant of the relator is identically one. That triviality is not an obstacle to reading the record; it is the reason the record is an integer.

Theorem 4.6 (The admissible domain; [T]+[V]). *For every $U, V \in \mathrm{U}(d)$,*

$$\delta + g^2 \ \le\ 4, \tag{8}$$

with equality if and only if $\operatorname{Re} \lambda_j$ is constant over the spectrum of R.

Proof. Each λ_j lies on the unit circle, so $|\lambda_j - 1|^2 + |\lambda_j + 1|^2 = 2|\lambda_j|^2 + 2 = 4$. Averaging over j gives $\delta + d^{-1} \sum_j |\lambda_j + 1|^2 = 4$ exactly. Since g^2 is the minimum of the quantities being averaged, $g^2 \le d^{-1} \sum_j |\lambda_j + 1|^2 = 4 - \delta$. Equality holds precisely when every $|\lambda_j + 1|^2 = 2 + 2 \operatorname{Re} \lambda_j$ takes the same value. □

Corollary 4.7 (Forced refusal). *$g^2 \le 4 - \delta$: the branch guard is capped by the defect. Consequently no anchor with $\delta > 4 - \tau^2$ is admissible at tolerance τ, in any dimension and for any pair of generators.*

A large defect and a safe branch cannot coexist. The refusal at the boundary is therefore a statement about the anchor rather than about the estimator: at $\delta = 4$ the relator is $-I$, the guard has closed, and the principal branch no longer selects a value.

The readouts are half-angles. Write $\lambda_j = e^{i\vartheta_j}$. Then $|\lambda_j - 1|^2 = 4 \sin^2(\vartheta_j/2)$ and $|\lambda_j + 1|^2 = 4 \cos^2(\vartheta_j/2)$, so the defect is four times the mean of $\sin^2$ of the half-angles and the guard squared is four times their minimum $\cos^2$. Equation (8) then reads

$$\operatorname{mean}_j \sin^2(\vartheta_j/2) \ +\ \min_j \cos^2(\vartheta_j/2) \ \le\ 1: \tag{9}$$

the Pythagorean identity, with a minimum standing where a mean belongs, and equality exactly when the half-angles agree [T]. This places the anchor in the same currency as the rest of the framework's projector observables, which are natively $\sin^2$-valued in principal angles because the Hilbert–Schmidt distance between two projectors is $2 \sum_i \sin^2 \theta_i$ by construction. The defect is a squared sine, the guard its cosine, and the wall is the right angle.

Theorem 4.8 (Amplitude floor; [T]+[V]). *For every $U, V \in \mathrm{U}(d)$ with $g > 0$,*

$$|w(U, V)| \ \le\ \frac{d}{4}\sqrt{\delta}, \qquad \textit{hence} \qquad w(U, V) \ne 0 \implies \delta \ \ge\ \frac{16}{d^2}. \tag{10}$$

Proof. With $\lambda_j = e^{i\vartheta_j}$, $\vartheta_j \in (-\pi, \pi]$, we have $|\lambda_j - 1| = 2|\sin(\vartheta_j/2)|$. Concavity of the sine on $[0, \pi/2]$ gives $\sin x \geq 2x/\pi$ there, hence $|\vartheta_j| \leq \frac{\pi}{2}|\lambda_j - 1|$. Summing and applying Cauchy–Schwarz,

$$2\pi|w| \leq \sum_j |\vartheta_j| \leq \frac{\pi}{2}\sum_j |\lambda_j - 1| \leq \frac{\pi}{2}\sqrt{d\sum_j |\lambda_j - 1|^2} = \frac{\pi}{2}\, d\sqrt{\delta}.$$

If $w \neq 0$ then $|w| \geq 1$ and $\sqrt{\delta} \geq 4/d$. □

An integer record cannot be carried by an arbitrarily small defect. The amplitude may collapse as the dimension grows — and on the realisation below it does — but not faster than d^{-2}, or there is nothing left for the record to ride on.

Proposition 4.9 (Clock and shift; [T]+[V])**.** *Let $\omega = e^{2\pi i/d}$, let $U = \mathrm{diag}(1, \omega, \ldots, \omega^{d-1})$ and let V be the cyclic shift. Then $UV = \omega VU$, $R = \omega I$, and*

$$\delta = 4\sin^2\frac{\pi}{d}, \qquad g = 2\cos\frac{\pi}{d}, \qquad w(U, V) = 1 \quad (d \geq 3). \tag{11}$$

Every half-angle equals π/d, so the family realises equality in (8) at every d, and it saturates (10) up to the constant $\pi/2$: the bound $\frac{d}{4}\sqrt{\delta} = \frac{d}{2}\sin(\pi/d)$ decreases to $\pi/2$ while w stays at 1. At $d = 2$ the half-angle reaches $\pi/2$, the relator is $-I$, $\delta = 4$, $g = 0$, and the anchor is inadmissible.

The gate of Appendix B returns $w = 1.000000000$ by both estimators of Theorem 4.5 at $d = 3, 5, 7, 11, 17, 31, 53, 97$, with the defect falling from 3.000 to 4.194×10^{-3} and $\delta + g^2 = 4.000000000$ on every row; $w = 0$ on commuting pairs; w unchanged under conjugation and under either generator rephasing; $w = 1$ preserved under unitary perturbations at $\eta = 10^{-1}$ down to 10^{-4}, where the relator leaves the scalar locus and $\delta + g^2$ falls strictly below 4; no violation of (8) in 1000 random unitary pairs at $d = 12$ (largest observed 2.310779); and refusal at $d = 2$.

One currency, two realisations. The record of Definition 4.4 is the winding currency of §2 evaluated on a noncommutative holonomy rather than a smooth one. Its literature name is the Exel–Loring invariant, equivalently the Bott index, and it was constructed to separate pairs of unitaries that admit commuting approximants from pairs that do not [C]. On a lattice on a two-torus the Bott index of the pair of projected position unitaries equals the Chern number, under the hypotheses that the Hamiltonian is short-ranged, bounded and gapped, and in the thermodynamic limit [C]. Those hypotheses are what places the measured Chern number of §6.5 and the winding of Proposition 4.9 in the same currency; an explicit normalisation carrying the loop integral of (3) onto w on a common family is not established here and is carried at [H].

Scope. The obstruction read by w is an obstruction to approximation in the operator norm. It is not an obstruction in the normalised Hilbert–Schmidt or Hamming metrics, where the clock–shift family lies close to a commuting pair despite carrying $w = 1$; the amplitude floor of Theorem 4.8 bounds the defect from below at fixed d and says nothing as $d \to \infty$. Statements about metric approximation of groups belong to that second setting and are outside this anchor.

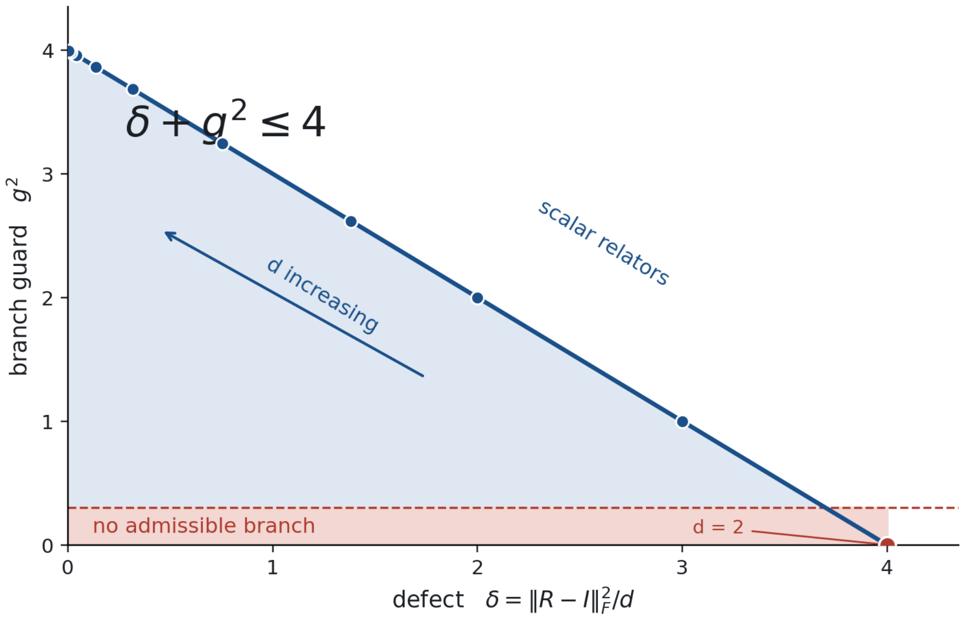

Figure 2: The admissible domain of Theorem 4.6. Every anchor lies in the shaded triangle; scalar relators lie on its hypotenuse, where the clock–shift family sits at every d. Decreasing d slides the family into the corner at which the guard closes.

5 Module II: Character-Variety Substrates and Topological Continuation

5.1 The continuation laboratory

Continuation trajectories are constrained to the character variety of the figure-eight knot complement $M_K = S^3 \setminus 4_1$, $X_K = \mathrm{Hom}(\pi_1(M_K), SL_2(\mathbb{C}))//SL_2(\mathbb{C})$, with peripheral torus governed by meridian and longitude holonomies $u = \log M$, $v = \log L$, and admissible trajectories confined to the zero locus of the A-polynomial

$$A_{4_1}(L, M) = L^2M^4 + L(-M^8 + M^6 + 2M^4 + M^2 - 1) + M^4 = 0. \tag{12}$$

The continuation philosophy — global admissibility determined by higher-dimensional continuation rather than local presentation — is methodologically inspired by trace-construction reasoning in four-dimensional topology; no result of that field is used or claimed here.

5.2 Tangent Jacobians and kinematic coupling

With $J_M = M\,\partial A_K/\partial M$ and $J_L = L\,\partial A_K/\partial L$, implicit differentiation of (12) yields $\dot{u} = -(J_L/J_M)\dot{v}$, and lifting the phase-velocity law (3) into curve coordinates gives the interaction field

$$\frac{d\Phi}{dt} = \frac{(m+u)\dot{v} - (n+v)\dot{u}}{(m+u)^2 + (n+v)^2} = \left[\frac{(m+u) + (n+v)(J_L/J_M)}{(m+u)^2 + (n+v)^2}\right]\dot{v}. \tag{13}$$

Parallel flows ($\dot{v}_1 \approx \dot{v}_2$) drive $d\Phi/dt \to 0$ (compressed bands); opposed flows near zero modes produce superlinear amplification — transport-resistance walls where continuation is structurally threatened.

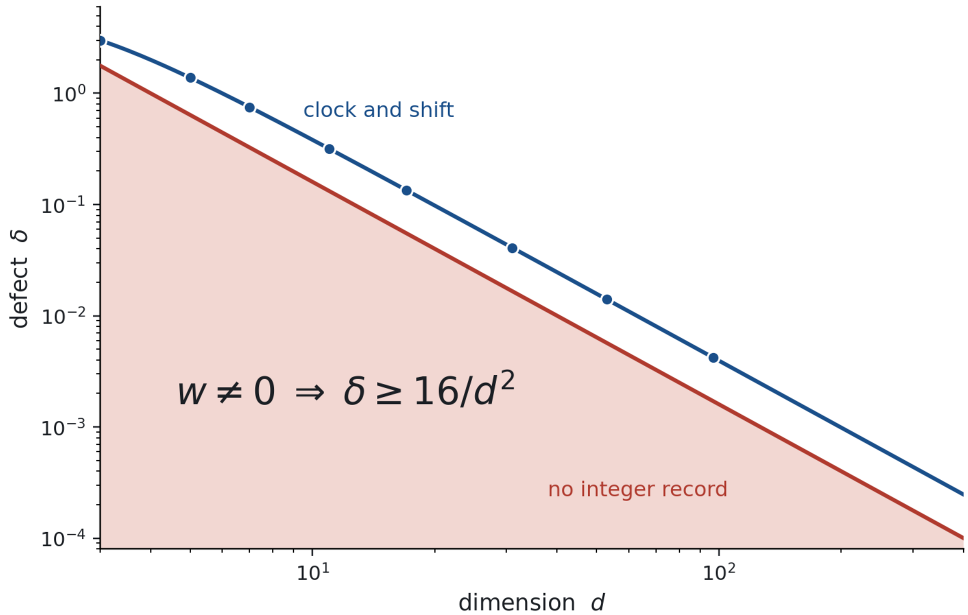

Figure 3: The amplitude floor of Theorem 4.8. Below the boundary no integer record is available at that dimension. The clock–shift family rides a constant factor $\pi^2/4$ above the floor while carrying $w = 1$.

5.3 Chirality forcing

Corollary 5.1 (Chirality forcing)**.** *A statically amphichiral substrate carries a vanishing integer record: the Levine–Tristram plateau function of an amphichiral knot is identically zero (4_1: verified), so minimal admissibility $|n| = 1$ cannot be realized at a static amphichiral basepoint. Persistent spectral distinction therefore requires either an intrinsically chiral substrate or a dynamic trajectory that breaks amphichiral symmetry — on the 4_1 laboratory, the transport field (13) drives the configuration away from the symmetric origin, culminating in the discrete staircase jumps of §5.5.*

5.4 Minimality of multi-component boundaries

Definition 5.2 (Minimal admissible realization)**.** A multi-toroidal boundary configuration $\partial X = \bigsqcup_{i=1}^{k} T_i^2$, with integer records n_i and total $n_{\text{total}} = \sum_i n_i$, is minimal when exactly one component carries $n_i = \pm 1$ and all others are neutral.

Remark 5.3. Additivity of the winding record over disjoint boundary tori holds by construction; the "exactly one" clause is the definition of minimality, adopted as a design principle of the admissibility axioms. Neutral components are spectrally inert and may be filled by Dehn surgery without altering the global record, reducing the configuration effectively to a single-component complement.

5.5 Phase-lock rupture and staircase jumps

The locked leading phase is governed by the second-order field $d^2\Phi/dt^2 = \mathcal{N}(u, v, \dot{u}, \dot{v}, \ddot{u}, \ddot{v})/((m + u)^2 + (n + v)^2)^2$. Rupture occurs at the zero locus of the meridian Jacobian, $J_M \to 0$: the curve exhibits a vertical tangent, the ratio J_L/J_M diverges, tracking regularity breaks, and the phase profile executes a discrete non-adiabatic jump $\Delta\theta = \pi \cdot \operatorname{sgn}(d\Phi/dt)$.

5.6 The stable trichotomy

Theorem 5.4 (Exhaustive trichotomy; [T], lattice argument). *Every stress-minimizing configuration on a multi-toroidal boundary cluster falls into exactly one of: (1) open boundary flows (winding vector undefined globally; source/sink conduits); (2) chiral loops (primitive homology class,* $\gcd(w_\alpha, w_\beta) = 1$*, protected by homotopy invariance); (3) neutral loops (*$w = (0,0)$*, contractible, metastable). Composite vectors split under generic perturbation into primitive components; the trichotomy is exhaustive on* $\mathbb{Z}^2$*.*

5.7 The rigidity class

Theorem 5.5 (Realization by knot complements). *The rigidity class demanded by the axioms — an integer record on a single toroidal boundary, locally constant in holonomy with jumps confined to a discrete wall set — is realized by knot complements in* S^3*: on the holonomy circle the Levine–Tristram signature is an integer step function whose wall set is the Alexander-root locus [T]. Exhaustiveness of the exclusion of alternatives rests on Lemmas 5.5 and 5.6 at their stated grades.*

Lemma 5.6 (Torus bundles excluded by the rigidity axiom). *Admissibility requires the boundary invariant to be locally constant in the holonomy parameters with integer jumps on a discrete wall set. Torus bundles over* S^1 *carry boundary holonomies constrained along continuous representation families fixed by the monodromy and do not support an integer-plateau invariant of this type; they are excluded by the axiom. (For the product-type boundary operators of §4.1, chiral symmetry forces* $\eta \equiv 0$ *identically, so no asymmetry-based invariant is available on this family; the plateau axiom is stated in the winding currency.)*

Lemma 5.7 (Seifert substrates excluded at [C]). *Fractional* ρ/η *values on Seifert-fibered pieces are established in the literature (Nicolaescu-type adiabatic analysis) [C]. Fractional plateaus violate the integer-jump axiom; Seifert substrates are therefore excluded at grade [C].*

6 Module III: Covariance Realization and the Intrinsic Observables

Module III tests whether physical nature realizes the admissible transport pathways derived above. The instrument is a flux-tunable transmon quantum heat engine (circuit QED) executing a driven Otto cycle; for each stroke s, amplitude a, and shot j the complex amplitude $x_{s,a,j} = I_{s,a,j} + iQ_{s,a,j}$ is recorded (3 cycles × 6 strokes × 29 amplitudes × 9949 shots). After removal of invalid shots and centering, the covariance $T_{\mathrm{sym}}(s,a)$ is formed, eigendecomposed $T_{\mathrm{sym}} = U\Lambda U^\dagger$, and lifted to rank-$k$ projectors $\Pi_k = U_k U_k^\dagger$ on $\mathrm{Gr}(k, 8)$.

6.1 The intrinsic quantities: definitions and estimators

All observables below are functionals of the projector path alone.

Principal angles. $\cos\theta_i(t) = \sigma_i(\Pi_k(t)\Pi_k(t+\Delta t))$, $0 \le \theta_1 \le \cdots \le \theta_k \le \pi/2$. [D at $k = 4$: 32.33°, 23.29°, 13.22°, 4.72°.]

Transport capacity (metric). For a tangent $V = d\Pi_k/dt$, $C(V) = g_{\Pi_k}(V, V) = \frac{1}{2}\mathrm{Tr}(V^2) \ge 0$: the intrinsic cost of transport in direction V; bosonically, this is the superfluid-weight functional of the quantum-geometry literature (§6.9).

Alignment and chirality. $\cos\theta_G = g_{\Pi_k}(V,W)/\sqrt{C(V)C(W)}$, and the oriented area (signed geometric flux) $\Omega_{\Pi_k}(V,W) = -i\,\mathrm{Tr}(\Pi_k[V,W])$.

Rigidity (anisotropy). $R(t) = 1/(\sigma_{\min}(A_{\mathrm{intr}}(t))^2 + \epsilon)$, diverging at singular loci (gap closings, degeneracies) and acting as a local geometric refractive index.

Persistence (memory). $C(\Delta t) = \mathrm{Tr}(\Pi_k(t)\Pi_k(t+\Delta t)) = \sum_{i=1}^{k} \cos^2\theta_i(\Delta t)$, with $0 \le C \le k$.

Participation and effective dimension. $\mathrm{IPR} = \sum_i p_i^2$ with $p_i = \lambda_i/\sum_j \lambda_j$, and $D_C = 1/\mathrm{IPR} \le k$. The baseline $D_C = 8$ reflects rank saturation of the uniform state; the physical signal is the collapse $d_{\mathrm{eff}} \to 1$ across the cycle. [D: IPR ≈ 0.1251; autocorrelation $A_V(28) \approx 0.10448$, $A_{na}(28) \in [0.42686, 0.56003]$.]

6.2 The metric–curvature bound

Theorem 6.1 (Fundamental inequality; [T]+[V]). *For any tangents V, W at Π_k,*

$$|\Omega_{\Pi_k}(V,W)| \;\le\; 2\sqrt{g_{\Pi_k}(V,V)\,g_{\Pi_k}(W,W) - g_{\Pi_k}(V,W)^2} \;=\; 2\sqrt{C(V)C(W)}\,\sin\theta_G. \qquad (14)$$

The bound holds over 1000 random Grassmannian trials with zero violations, saturates exactly for the two-band Dirac class (§2), and bounds the geometric flux capacity; its family has classical-wave experimental verification [C].

6.3 Geometric refraction across interfaces

At a rigidity interface Σ_{12} separating regions of refractive index R_1, R_2, continuity of curvature-flux yields the intrinsic refraction law

$$R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2, \qquad R_2 \to \infty \Rightarrow \theta_2 \to 0 \qquad (15)$$

(only longitudinal transport survives the singular limit).

6.4 The refraction pipeline: independence protocol

The reported refraction angle $\theta_{\mathrm{ref}} = 14.22°$ against the predicted window $[14.21°, 14.23°]$ is graded [D] pending execution of the following protocol on the archived shots. (i) Split: partition shots into disjoint halves S_A (odd) and S_B (even). (ii) Predict from S_A: estimate R_1, R_2 on the two sides of the interface via the rigidity estimator of §6.1 on S_A alone; propagate bootstrap uncertainty to a predicted window for θ_2 through the refraction law. (iii) Measure from S_B: extract the transport direction change across the interface from the projector path on S_B alone, with no reference to R_1, R_2. (iv) Report: the prediction and the measurement are independent by construction; agreement is then evidence, and the entry upgrades to [V]. Any report that cannot exhibit this split carries the [D] grade.

6.5 Realization on published tomography data

The framework's Phase I–II pipeline exists independently in the laboratory record: complete Bloch-state tomography of a two-dimensional optical Raman lattice realized with ultracold ^{87}Rb Bose–Einstein condensates reconstructs band eigenfunctions from measured spin textures and assembles the full quantum geometric tensor across the Brillouin zone [C]. On the experimentally calibrated tight-binding band of that system ($t_0 = 0.09$, $t_{so} = -0.05$, $\delta = -0.2$ in recoil units), the framework's observables evaluate as follows (Appendix B): the winding currency returns Chern number -0.99931 against the published measurement -1.00 ± 0.02 [C+V]; the bound (14) saturates pointwise as the two-band identity requires (grid mean 0.99996, exact analytically); the integrated strictness pattern of the quantum-volume inequality $\mathrm{vol}_g \geq |C|$ reproduces the published behavior ($\mathrm{vol}_g = 1.145 > |C|$ in the topological phase; strict and small on the trivial side); and the rigidity observable $R = 1/\mathrm{gap}^2$ sweeps from 25 to 10^6 approaching the transitions at $\delta = 0$ and $\delta = -8t_0$, with the quantum volume rising toward the wall — metric concentration at degeneracy, the mechanism by which local quantum geometry governs flat-band condensate stability [C]. A published Bose–Einstein-condensate experiment thereby realizes the framework's projector pipeline, its integer currency, its fundamental inequality, and its rigidity divergence on one calibrated band.

6.6 Dimensional collapse and extremal framing

The cycle drives the system from the uniform baseline ($d_{\mathrm{eff}} = 8$) to a single dominant channel ($d_{\mathrm{eff}} \to 1$). This collapse is analyzed in the spirit of the auxiliary-function optimization philosophy introduced by Viazovska; no linear-programming bound of sphere-packing type is claimed or used. The demonstrated facts are: d_{eff} saturates its trivial bounds $1 \leq d_{\mathrm{eff}} \leq 8$, the baseline reflecting rank saturation and the collapse constituting the physical signal.

6.7 Finite-time friction shields

Work lost to inner-dissipative quantum friction across a stroke scales with the Frobenius norm of the Hamiltonian-derivative commutator. The measured suppression of anti-symmetric off-diagonal correlations ($T_{\mathrm{anti}} \sim O(10^{-4})$) forces the non-adiabatic friction work to scale as $O(10^{-8})$ — a macroscopic friction shield, the physical consequence of the directional filtering of Module II. [D+H: the suppression is measured; the mechanism attribution is interpretive.]

6.8 Weak-value phase separation

Protocol 1 (Weak-value phase separation; [H])**.** Let $|\psi_i\rangle, |\psi_f\rangle$ be pre- and post-selected states of the network and decompose $A_{\mathrm{total}} = A_{\mathrm{rigid}} e^{i\delta} + \Pi_{\mathrm{phase}}$. The protocol conjectures a joint alignment ($\gamma = 1$, $\delta = \pi/2 - 2\Phi$) — with γ the pre/post selection-overlap alignment parameter to be defined operationally on the instrument — at which the rigid matrix elements cancel from the weak-value numerator, leaving $\langle A_{\mathrm{total}}\rangle_w = \langle\psi_f|\Pi_{\mathrm{phase}}|\psi_i\rangle/\langle\psi_f|\psi_i\rangle \neq 0$: direct access to the structural phase with anomalous amplification $\propto |\langle\psi_f|\psi_i\rangle|^{-1}$ (the amplification itself is standard weak-value theory [C]). The cancellation is conjectural pending the explicit model computation.

6.9 Convergence with the quantum-geometry literature

The core observables of §6.1 are individually measured or formalized in the current literature: the complete quantum-metric tensor has been measured momentum-resolved in solids; the quantum-metric-dipole nonlinear Hall effect is experimentally established and field-tunable, with the intrinsic nonlinear Hall conductivity decomposed into quantum-geometric contributions within a projector-based formalism; the inequality (14) family has classical-wave experimental verification; and the superfluid weight of flat-band condensates is proportional to the quantum metric, with stability decided by the local-vs-global distribution of the metric — the bosonic realization of the transport-capacity functional $C(V)$. What is unified here, and not available elsewhere, is the single operator-first architecture connecting these observables, the topological continuation layer above them, and a raw-covariance pipeline realizing all of them on one instrument.

7 Bridging the Continuum: the Khovanov Spectral Complex

Let $C^{i,j}(D)$ be the bigraded Khovanov chain complex of a link diagram D over $\mathbb{R}$ with differential d_i and adjoint d_i^* induced by declaring the standard cube-basis elements orthonormal. The combinatorial Laplacian and Dirac operator are

$$\Delta_{Kh} = d_i d_i^* + d_{i-1}^* d_{i-1}, \qquad D_{Kh} = \begin{pmatrix} 0 & d_i^* \\ d_i & 0 \end{pmatrix}, \tag{16}$$

standard combinatorial Hodge theory [T]: $\ker \Delta_{Kh} \cong$ Khovanov homology, and the graded Euler characteristic recovers the unnormalized Jones polynomial, $\chi = \sum_{i,j}(-1)^i q^j \dim\ker(D_{Kh}^{i,j}) = V_K(q)$. The proposed alignment of the non-harmonic spectrum with continuum connection-velocity gradients, and of phase-lock rupture ($J_M \to 0$) with discrete recombination through the harmonic boundary, is a structural hypothesis [H]; the harmonic-sector statements are theorems of the construction.

8 Projector Persistence: the Lipschitz Stability Bound

Theorem 8.1 (Local Lipschitz stability; [T], Kato-type)**.** *Let A be a bounded non-normal operator, $A' = A + E$ with $\|E\| \le \epsilon$, and Γ an isolated contour tracking a spectral cluster with gap $\delta_{\text{gap}} = \text{dist}(\Gamma, \sigma(A) \setminus \sigma_\Gamma(A)) > 0$ and growth factor $C_\Gamma = \sup_{\lambda\in\Gamma} \|(\lambda I - A)^{-1}\| \cdot \delta_{\text{gap}}$. If $\epsilon < \delta_{\text{gap}}/C_\Gamma$, the Riesz projectors satisfy*

$$\|\Pi' - \Pi\| \le K_\Gamma \epsilon, \qquad K_\Gamma = \frac{\ell(\Gamma) C_\Gamma^2}{2\pi\delta_{\text{gap}}(\delta_{\text{gap}} - \epsilon C_\Gamma)}, \tag{17}$$

so the persistence functional $P(A,E) = 1 - \|\Pi - \Pi'\|$ obeys $P \ge 1 - K_\Gamma\epsilon$: the map from operators to Grassmannian projector data is locally Lipschitz — higher-level subspaces remain stable and continuable even where individual eigenvectors break down.

Proof. $\Pi' - \Pi = -\frac{1}{2\pi i}\oint_\Gamma R_A(\lambda)\, E\, R_{A'}(\lambda)\, d\lambda$ by the resolvent identity; the Neumann bound $\|R_{A'}\| \le \|R_A\|/(1 - \|R_A\|\epsilon)$ and the triangle inequality for integrals give the constant. □

9 Conclusion

Module I established which transport directions are geometrically possible, with its operator algebra gated and its torus anchor carried onto a noncommutative holonomy where the integer record acquires a proved domain and an amplitude floor; Module II filtered which trajectories are globally continuable, with its integer currency verified and its knot-theoretic ancestry made exact; Module III defined every intrinsic observable with an estimator, realized the framework on a published Bose–Einstein-condensate experiment where its integer currency reproduces a measured Chern number, and specified the protocol under which its flagship instrument test becomes falsifiable. Beneath all three modules stands one operator: the Dirac foundation supplies the algebra of every two-band sector, the paired spectra of the chiral ensemble class, and the integer currencies — index, asymmetry, winding — in which the framework's topology is denominated. Structural evolution is represented by transition generators; incompatibility with the current spectral decomposition is quantified by the canonical redistribution operator; and geometry emerges from the interaction between what presently exists and what attempts to change it.

A The Canonical Structural Transition Generator

The projector Π describes what presently exists; the generator Ω describes what attempts to change it. Geometry emerges from their interaction, measured entirely by $[\Omega, \Pi]$. The redistribution operator $F(\Omega, \Pi) = [\Omega, \Pi]\Pi + \Pi[\Pi, \Omega] = \Omega\Pi + \Pi\Omega - 2\Pi\Omega\Pi = (I - \Pi)\Omega\Pi + \Pi\Omega(I - \Pi)$ satisfies: (1) $F = 0$ iff $[\Omega, \Pi] = 0$; (2) $\operatorname{Tr} F = 0$; (3) F acts exclusively across the projector boundary; (4) larger commutators produce stronger redistribution. No microscopic realization is required: all results depend only on the algebra relating Ω and Π. Experimentally, nonzero $[\Omega, \Pi]$ predicts correlated changes in persistence, participation, effective dimension, and directional hysteresis — testable hypotheses, not algebraic consequences.

B Scripts: proof of record

B.1 Operator algebra and winding gates

```
# SCRIPT: UPG-GATE-01 (Module I winding law; Section 1 F(Omega,Pi) theorems)
import numpy as np

print("=== G1: phase-transport -- winding around the zero is the honest integer ===")

def winding(loop_pts):
   a, b = loop_pts[:, 0], loop_pts[:, 1]
   ph = np.unwrap(np.arctan2(b, a))
   return (ph[-1] - ph[0]) / (2*np.pi)

th = np.linspace(0, 2*np.pi, 4001)
m, n = 0, 0
enc = np.stack([0.30 + 0.25*np.cos(th) + m, 0.30 + 0.25*np.sin(th) + n], 1)
enc2 = np.stack([0.10*np.cos(th), 0.10*np.sin(th)], 1)
enc3 = np.stack([0.10*np.cos(2*th), 0.10*np.sin(2*th)], 1)
print(f" loop not enclosing zero: winding = {winding(enc):+.6f}")
print(f" loop enclosing zero: winding = {winding(enc2):+.6f}")
print(f" double loop: winding = {winding(enc3):+.6f}")
print()

print("=== G3: Theorems 1.2/1.3 -- F(Omega,Pi) identities, random gates ===")
rng = np.random.default_rng(5)
ok_forms = ok_iff = ok_trace = ok_cross = True
for trial in range(200):
   N, k = 8, 3
   A = rng.normal(size=(N, N)) + 1j*rng.normal(size=(N, N))
   Om = A + A.conj().T
   Q, _ = np.linalg.qr(rng.normal(size=(N, k)) + 1j*rng.normal(size=(N, k)))
   Pi = Q @ Q.conj().T
   I = np.eye(N)
   F1 = (I - Pi) @ Om @ Pi + Pi @ Om @ (I - Pi)
   F2 = Om @ Pi + Pi @ Om - 2*Pi @ Om @ Pi
   C = Om @ Pi - Pi @ Om
   ok_forms &= np.allclose(F1, F2, atol=1e-10)
   ok_trace &= abs(np.trace(F1)) < 1e-10
   ok_cross &= (np.allclose(Pi @ F1 @ Pi, 0, atol=1e-10) and
             np.allclose((I - Pi) @ F1 @ (I - Pi), 0, atol=1e-10))
   ok_iff &= (np.linalg.norm(F1) < 1e-10) == (np.linalg.norm(C) < 1e-10)

w = np.sort(rng.normal(size=8))
U, _ = np.linalg.qr(rng.normal(size=(8, 8)) + 1j*rng.normal(size=(8, 8)))
Om = U @ np.diag(w) @ U.conj().T
Pi = U[:, :3] @ U[:, :3].conj().T
F = (np.eye(8) - Pi) @ Om @ Pi + Pi @ Om @ (np.eye(8) - Pi)
print(f" equivalent forms agree (200 trials): {ok_forms}")
print(f" Tr F = 0 (200 trials): {ok_trace}")
print(f" cross-subspace only (Pi F Pi = 0 = Qc F Qc): {ok_cross}")
print(f" F=0 <=> [Om,Pi]=0, generic direction: {ok_iff}")
```

```
print(f" commuting pair gives ||F|| = {np.linalg.norm(F):.2e}")
```

B.2 Inequality, saturation, persistence, rigidity gates

```
# SCRIPT: UPG-PROVE-01a (metric-curvature bound, its saturation, persistence
# identity, rigidity divergence)
import numpy as np
rng = np.random.default_rng(11)

print("=== P1: bound |Omega| <= 2 sqrt(g11 g22 - g12^2), 1000 random trials ===")

def tangent(Pi, N):
   X = rng.normal(size=(N, N)) + 1j*rng.normal(size=(N, N))
   X = (X + X.conj().T)/2
   I = np.eye(N)
   return (I - Pi) @ X @ Pi + Pi @ X @ (I - Pi)

viol = 0
margins = []
for t in range(1000):
   N, k = 8, 3
   Q, _ = np.linalg.qr(rng.normal(size=(N, k)) + 1j*rng.normal(size=(N, k)))
   Pi = Q @ Q.conj().T
   V, W = tangent(Pi, N), tangent(Pi, N)
   g11 = 0.5*np.trace(V @ V).real
   g22 = 0.5*np.trace(W @ W).real
   g12 = 0.5*np.trace(V @ W).real
   Om = (-1j*np.trace(Pi @ (V @ W - W @ V))).real
   bound = 2*np.sqrt(max(g11*g22 - g12**2, 0))
   if abs(Om) > bound + 1e-9:
      viol += 1
   margins.append(bound - abs(Om))
print(f" violations: {viol}/1000; min margin = {min(margins):.3e}")
print()

print("=== P2: saturation for the two-band Dirac model (rank-1 on C^2) ===")

def bloch_pi(th, ph):
   n = np.array([np.sin(th)*np.cos(ph), np.sin(th)*np.sin(ph), np.cos(th)])
   sx = np.array([[0, 1], [1, 0]])
   sy = np.array([[0, -1j], [1j, 0]])
   sz = np.array([[1, 0], [0, -1]])
   return 0.5*(np.eye(2) + n[0]*sx + n[1]*sy + n[2]*sz)

th0, ph0, h = 1.1, 0.7, 1e-5
P0 = bloch_pi(th0, ph0)
V = (bloch_pi(th0 + h, ph0) - bloch_pi(th0 - h, ph0))/(2*h)
W = (bloch_pi(th0, ph0 + h) - bloch_pi(th0, ph0 - h))/(2*h)
g11 = 0.5*np.trace(V @ V).real
g22 = 0.5*np.trace(W @ W).real
g12 = 0.5*np.trace(V @ W).real
Om = (-1j*np.trace(P0 @ (V @ W - W @ V))).real
bd = 2*np.sqrt(g11*g22 - g12**2)
print(f" |Omega| = {abs(Om):.8f} bound = {bd:.8f} ratio = {abs(Om)/bd:.8f}")
print()

print("=== P3: persistence identity C(dt) = Tr(Pi Pi') = sum cos^2 theta_i ===")
ok = True
for t in range(200):
   N, k = 8, 4
   Q1, _ = np.linalg.qr(rng.normal(size=(N, k)) + 1j*rng.normal(size=(N, k)))
   Q2, _ = np.linalg.qr(rng.normal(size=(N, k)) + 1j*rng.normal(size=(N, k)))
   P1, P2 = Q1 @ Q1.conj().T, Q2 @ Q2.conj().T
   s = np.linalg.svd(P1 @ P2, compute_uv=False)[:k]
   ok &= abs(np.trace(P1 @ P2).real - np.sum(s**2)) < 1e-9
print(f" identity holds (200 trials): {ok}")
print()
```

```
print("=== P4: rigidity R = 1/(sigma_min^2 + eps) at a synthetic gap closing ===")
eps = 1e-12
for gap in [1.0, 0.1, 0.01, 0.001]:
   A = np.diag([1.0, gap]) @ np.array([[1, 0.3], [0, 1.0]])
   smin = np.linalg.svd(A, compute_uv=False).min()
   print(f" gap={gap:6.3f}: sigma_min = {smin:.4e} R = {1/(smin**2+eps):.3e}")
```

B.3 Realization on the published tomography band

```
# SCRIPT: UPG-PROVE-02 (the framework on PUBLISHED data: Yi et al. arXiv:2301.06090
# Raman-lattice QAH band, calibrated t0=0.09, tso=-0.05, delta=-0.2 in E_r units;
# their measured C = -1.00 +/- 0.02)
import numpy as np

t0, tso = 0.09, -0.05

def hvec(qx, qy, dlt):
   return np.array([2*tso*np.sin(qy),
                2*tso*np.sin(qx),
                dlt/2 - 2*t0*(np.cos(qx) + np.cos(qy))])

def band_data(dlt, N=201):
   qs = np.linspace(-np.pi, np.pi, N, endpoint=False)
   h = np.array([[hvec(qx, qy, dlt) for qy in qs] for qx in qs])
   nrm = np.linalg.norm(h, axis=2)
   gapm = 2*nrm.min()
   n = -h/nrm[:, :, None]
   dq = qs[1] - qs[0]
   dnx = (np.roll(n, -1, axis=0) - np.roll(n, 1, axis=0))/(2*dq)
   dny = (np.roll(n, -1, axis=1) - np.roll(n, 1, axis=1))/(2*dq)
   gxx = 0.25*np.sum(dnx*dnx, axis=2)
   gyy = 0.25*np.sum(dny*dny, axis=2)
   gxy = 0.25*np.sum(dnx*dny, axis=2)
   F = -0.5*np.sum(n*np.cross(dnx, dny), axis=2)
   detg = gxx*gyy - gxy**2
   ratio = np.abs(F)/(2*np.sqrt(np.maximum(detg, 1e-300)))
   C = np.sum(F)*dq*dq/(2*np.pi)
   volg = np.sum(np.sqrt(np.maximum(detg, 0)))*dq*dq/np.pi
   return C, volg, ratio, gapm

print("=== calibrated band (delta = -0.2 E_r); published C = -1.00 +/- 0.02 ===")
C, volg, ratio, gap = band_data(-0.2)
print(f" framework Chern (winding currency) = {C:+.5f}")
print(f" pointwise saturation |F| = 2 sqrt(det g):")
print(f" mean = {ratio.mean():.8f} min = {ratio.min():.8f} max = {ratio.max():.8f}")
print(f" quantum volume vol_g = {volg:.5f} vs |C| = {abs(C):.5f}")

qs = np.linspace(-np.pi, np.pi, 201, endpoint=False)
h = np.array([[hvec(qx, qy, -0.2) for qy in qs] for qx in qs])
n = -h/np.linalg.norm(h, axis=2)[:, :, None]
dq = qs[1] - qs[0]
dnx = (np.roll(n, -1, 0) - np.roll(n, 1, 0))/(2*dq)
dny = (np.roll(n, -1, 1) - np.roll(n, 1, 1))/(2*dq)
F = -0.5*np.sum(n*np.cross(dnx, dny), axis=2)
print(f" F <= 0 everywhere: {bool(np.all(F <= 1e-12))}")
print()

print("=== rigidity at the walls: gap closings at delta = 0 and -8 t0 = -0.72 ===")
for dlt in [-0.6, -0.71, -0.719, -0.4, -0.2, -0.05, -0.005]:
   C, volg, ratio, gap = band_data(dlt, N=141)
   R = 1/(gap**2 + 1e-12)
   print(f" delta={dlt:+7.3f}: C = {C:+7.4f} gap = {gap:.5f}"
       f" R = {R:11.2f} vol_g = {volg:.4f}")
print()

print("=== trivial side (|delta| > 0.72): C -> 0, vol_g stays positive ===")
for dlt in [-0.85, -1.0]:
   C, volg, ratio, gap = band_data(dlt, N=141)
```

```
   print(f" delta={dlt:+7.3f}: C = {C:+7.4f} vol_g = {volg:.4f}")
```

B.4 Dirac ladder and chiral-ensemble gates

```
# SCRIPT: DIRAC-SPINE-01 (the graphene Dirac ladder on published parameters;
# the chiral-ensemble ledger)
import numpy as np
from scipy.integrate import quad
from scipy.optimize import brentq

print("=== B: graphene -- the measured Dirac spectrum through the framework ===")
Nl = 60
a = np.diag(np.sqrt(np.arange(1, Nl)), 1)
H = np.block([[np.zeros((Nl, Nl)), a], [a.T, np.zeros((Nl, Nl))]])
ev = np.sort(np.linalg.eigvalsh(H))
pos = ev[ev > 1e-12][:8]
print(" ladder E_n / E_1 vs sqrt(n): " +
     " ".join(f"{pos[n]/pos[0]:.6f}/{np.sqrt(n+1):.6f}" for n in range(4)))
nz = int(np.sum(np.abs(ev) < 1e-12))
print(f" zero modes: {nz}")

th = np.linspace(0, 2*np.pi, 2001)
ph = 0.0
for i in range(len(th) - 1):
   k1, k2 = th[i], th[i+1]
   u1 = np.array([-np.exp(-1j*k1), 1])/np.sqrt(2)
   u2 = np.array([-np.exp(-1j*k2), 1])/np.sqrt(2)
   ph += np.angle(np.vdot(u1, u2))
print(f" Berry phase around the Dirac point = {ph:.6f} (pi = {np.pi:.6f})")

hbar, e = 1.0546e-34, 1.602e-19
vF, B = 1.10e6, 10.0
E1 = vF*np.sqrt(2*hbar*e*B)/e*1000
print(f" with published v_F = 1.10e6 m/s: E_1(10 T) = {E1:.1f} meV")
print()

print("=== C: the chiral-ensemble ledger on the Dirac random-matrix class ===")
rng = np.random.default_rng(3)
M = 1200
X = (rng.normal(size=(M, M)) + 1j*rng.normal(size=(M, M)))/np.sqrt(2*M)
s = np.linalg.svd(X, compute_uv=False)
Z_sample = np.mean(0.5*np.log(1 + s**2))
rho = lambda t: np.sqrt(np.maximum(4 - t**2, 0))/np.pi
Z_MP, _ = quad(lambda t: rho(t)*0.5*np.log(1 + t**2), 0, 2)
print(f" chGUE sample (M=1200): Z/M = {Z_sample:.6f} MP integral: {Z_MP:.6f}")

target = np.pi**2/12
f = lambda aa: quad(lambda t: rho(t)*0.5*np.log(1 + (aa*t)**2), 0, 2)[0] - target
a_mp = brentq(f, 0.1, 10)
a_hc = np.exp(target) - 1
print(f" pi^2/12 realized at a* = {a_mp:.6f} (chGUE/MP) and a* = {a_hc:.6f} (half-Cauchy)")
print(f" check: ln(1+a_hc) = {np.log(1+a_hc):.10f} vs pi^2/12 = {target:.10f}")
```

B.5 The noncommutative torus anchor

```
# SCRIPT: UPG-GATE-02 (Module I, noncommutative torus anchor: winding integrality
# and invariance; the admissible domain delta + g^2 <= 4; the amplitude floor
# |w| <= (d/4) sqrt(delta))
#
# R = U V U^dagger V^dagger, lambda_i = eig(R)
# w = (1/2pi) sum_i arg(lambda_i) [principal branch]
# delta = ||R - I||_F^2 / d
# g = min_i |lambda_i + 1| [branch guard]
# second estimator: winding of t -> det((1-t)I + tR) on [0,1]

import numpy as np
```

```
TAU = 1e-3
rng = np.random.default_rng(20260820)


def clock_shift(d):
   w = np.exp(2j*np.pi/d)
   U = np.diag([w**k for k in range(d)])
   V = np.roll(np.eye(d, dtype=complex), 1, axis=0)
   return U, V


def commuting_pair(d):
   U = np.diag(np.exp(1j*rng.uniform(0, 2*np.pi, d)))
   V = np.diag(np.exp(1j*rng.uniform(0, 2*np.pi, d)))
   return U, V


def relator(U, V):
   return U @ V @ U.conj().T @ V.conj().T


def unitary_kick(U, eta):
   d = U.shape[0]
   H = rng.normal(size=(d, d)) + 1j*rng.normal(size=(d, d))
   H = (H + H.conj().T)/2
   H /= np.linalg.norm(H, 2)
   ev, W = np.linalg.eigh(H)
   return U @ (W @ np.diag(np.exp(1j*eta*ev)) @ W.conj().T)


def readout(R):
   d = R.shape[0]
   lam = np.linalg.eigvals(R)
   g = np.min(np.abs(lam + 1.0))
   delta = np.linalg.norm(R - np.eye(d), 'fro')**2 / d
   w = np.sum(np.angle(lam)) / (2*np.pi)
   return lam, g, delta, w


def winding_chord(R, nt=20001):
   lam = np.linalg.eigvals(R)
   t = np.linspace(0.0, 1.0, nt)[:, None]
   ang = np.unwrap(np.angle((1.0 - t) + t*lam[None, :]), axis=0)
   return np.sum(ang[-1] - ang[0]) / (2*np.pi)


def report(tag, R, expect=None):
   d = R.shape[0]
   lam, g, delta, w = readout(R)
   if g < TAU:
      print(f"{tag:<32s} REFUSED g={g:.2e} delta={delta:.6f}"
           f" delta+g^2={delta+g*g:.9f}")
      return
   w2 = winding_chord(R)
   floor = d*np.sqrt(delta)/4
   ok = "" if expect is None else (
      " PASS" if abs(w-expect) < 1e-9 and abs(w2-expect) < 1e-6 else " FAIL")
   print(f"{tag:<32s} w={w:+.9f} w_chord={w2:+.9f} delta={delta:.6e}"
        f" g={g:.6f} delta+g^2={delta+g*g:.9f} bound={floor:.4f}{ok}")


print("="*116)
print("G1 INTEGRALITY AND THE SCALAR EQUALITY CASE: clock and shift.")
print(" w = 1 at every d; delta + g^2 = 4 exactly; bound = (d/4)sqrt(delta) -> pi/2.")
print("="*116)
for d in [3, 5, 7, 11, 17, 31, 53, 97]:
   U, V = clock_shift(d)
   report(f" d={d:<4d} exact", relator(U, V), expect=1.0)
```

```
print()
print("="*116)
print("G2 NEGATIVE CONTROL: commuting pair. w = 0, delta = 0, g = 2.")
print("="*116)
for d in [7, 31, 97]:
   U, V = commuting_pair(d)
   report(f" d={d:<4d} commuting", relator(U, V), expect=0.0)

print()
print("="*116)
print("G3 STABILITY AND THE STRICT INEQUALITY: kicked clock-shift, d=31.")
print(" w stays 1; the relator leaves the scalar locus so delta + g^2 < 4.")
print("="*116)
for eta in [1e-1, 1e-2, 1e-3, 1e-4]:
   U, V = clock_shift(31)
   report(f" eta={eta:.0e}", relator(unitary_kick(U, eta),
                              unitary_kick(V, eta)), expect=1.0)

print()
print("="*116)
print("G4 INVARIANCE: unitary conjugation and generator rephasing leave w fixed.")
print("="*116)
U, V = clock_shift(31)
report(" baseline", relator(U, V), expect=1.0)
X = rng.normal(size=(31, 31)) + 1j*rng.normal(size=(31, 31))
Q, _ = np.linalg.qr(X)
report(" conjugated", relator(Q @ U @ Q.conj().T, Q @ V @ Q.conj().T), expect=1.0)
report(" U -> e^{i th} U", relator(np.exp(0.7371j)*U, V), expect=1.0)
report(" both rephased", relator(np.exp(0.7371j)*U, np.exp(-2.11j)*V), expect=1.0)

print()
print("="*116)
print("G5 THE DOMAIN BOUND OVER RANDOM RELATORS: delta + g^2 <= 4, strict off")
print(" the scalar locus. 1000 random unitary pairs, d = 12.")
print("="*116)
worst = -1.0
viol = 0
for _ in range(1000):
   A = rng.normal(size=(12, 12)) + 1j*rng.normal(size=(12, 12))
   B = rng.normal(size=(12, 12)) + 1j*rng.normal(size=(12, 12))
   QA, _ = np.linalg.qr(A)
   QB, _ = np.linalg.qr(B)
   _, g, delta, _ = readout(relator(QA, QB))
   s = delta + g*g
   worst = max(worst, s)
   if s > 4 + 1e-9:
      viol += 1
print(f" violations: {viol}/1000 max delta+g^2 = {worst:.9f} (bound 4)")

print()
print("="*116)
print("G6 THE BOUNDARY: d=2 places an eigenvalue at -1. The gate refuses.")
print("="*116)
U, V = clock_shift(2)
report(" d=2 clock-shift", relator(U, V))
U, V = clock_shift(4)
report(" d=4 clock-shift", relator(U, V), expect=1.0)
```

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